Problem set 2 – answers

Twists and Wrenches

Modern Robotics

2013

1. $\dot{H}$

Suppose that you want to use a homogeneous matrix $H_{\mathrm{b}}^{0}(t) \in SE(3)$ to describe the orientation of a body (frame) Ψ_b with respect to some inertial frame Ψ_0—for instance in a simulation. Of course, in the simulation you can calculate the velocity of a body easily enough by integrating the force, using $\dot{v} = 1/m \cdot F \Rightarrow v = 1/m \int F$.[1] Doing this in a three-dimensional Euclidean space will give you the full 6-D twist.

Now for the question: how can you calculate $H_{\mathrm{b}}^{0}(t)$, given $H_{\mathrm{b}}^{0}(0)$ (the initial configuration) and the twist of the body expressed in its own frame, $T_{\mathrm{b}}^{\mathrm{b},0}$?

Hint: consider the definition of the tilde form of the twist $\tilde{T}_{b}^{b,0}$.

Starting from the definition[2] of $\tilde{T}_{\mathrm{b}}^{\mathrm{b},0}$:

$$\begin{aligned} \tilde{T}_{\mathrm{b}}^{\mathrm{b},0} &:= H_{0}^{\mathrm{b}}\ \dot{H}_{\mathrm{b}}^{0} \\ (H_{0}^{\mathrm{b}})^{-1}\ \tilde{T}_{\mathrm{b}}^{\mathrm{b},0} &= \dot{H}_{\mathrm{b}}^{0} \\ H_{\mathrm{b}}^{0}\ \tilde{T}_{\mathrm{b}}^{\mathrm{b},0} &= \dot{H}_{\mathrm{b}}^{0} \end{aligned}$$

So the orientation and position of a body b can be found as:

$$H_{\mathrm{b}}^{0}(t) = \int_{0}^{t} H_{\mathrm{b}}^{0}(\tau) \cdot \tilde{T}_{\mathrm{b}}^{\mathrm{b},0}(\tau)\ \mathrm{d}\tau + H_{\mathrm{b}}^{0}(0)$$

2. Tilde form of twists

Show that the left or right translation of velocity vectors $\dot{H}$ (for $H \in SE(3)$), that result in $\tilde{T}_{\mathrm{i}}^{\mathrm{i,j}}$ and $\tilde{T}_{\mathrm{i}}^{\mathrm{j,j}}$ respectively, indeed belong to

$$se(3) := \left\{ \begin{pmatrix} \Omega & v \\ \mathbf{0} & 0 \end{pmatrix} \text{ s.t. } \Omega \in so(3),\ v \in \mathbb{R}^3 \right\} \tag{1}$$

[1] You might know this relation as $F = ma$ instead.

[2] Remember the mnemonic: the frame the twist is expressed in, the left-most superscript of $\tilde{T}$, is the outer index of the H-matrices (top left, bottom right). The H-matrix that gets the dot is the one whose indices correspond to the other indices of $\tilde{T}$, i.e. b low and 0 high; motion of b w.r.t. 0.

Hint: look at how the right and left translations are defined.

For example, look at $\tilde{T}_{\mathrm{i}}^{\mathrm{j},\mathrm{j}}$:

$$\begin{aligned}
\tilde{T}_{\mathrm{i}}^{\mathrm{j},\mathrm{j}} &:= \dot{H}_{\mathrm{i}}^{\mathrm{j}} \cdot H_{\mathrm{j}}^{\mathrm{i}} \\
&= \begin{pmatrix} \dot{R}_{\mathrm{i}}^{\mathrm{j}} & \dot{p}_{\mathrm{i}}^{\mathrm{j}} \\ 0 & \dot{1} \end{pmatrix} \cdot \begin{pmatrix} R_{\mathrm{j}}^{\mathrm{i}} & p_{\mathrm{j}}^{\mathrm{i}} \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \dot{R}_{\mathrm{i}}^{\mathrm{j}} & \dot{p}_{\mathrm{i}}^{\mathrm{j}} \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} R_{\mathrm{i}}^{\mathrm{j}\top} & p_{\mathrm{j}}^{\mathrm{i}} \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \dot{R}_{\mathrm{i}}^{\mathrm{j}} R_{\mathrm{i}}^{\mathrm{j}\top} & \dot{R}_{\mathrm{i}}^{\mathrm{j}} p_{\mathrm{j}}^{\mathrm{i}} + \dot{p}_{\mathrm{i}}^{\mathrm{j}} \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} \Omega & v \\ 0 & 0 \end{pmatrix} \quad , \quad \Omega \in so(3), v \in \mathbb{R}^3
\end{aligned}$$

where in the last step Ω is in $so(3)$ because $R \in SO(3)$. $\square$

3. Fixed frames

Consider the two bodies of Figure 1, which move with respect to each other and both have two coordinate frames affixed to it. You can describe the relative motions of these two objects in various ways: $T_3^{1,1}, T_4^{2,2}, T_4^{1,1}$ for example all represent the motion of body B with respect to body A, expressed in one of the frames on A.

Try tro prove that $T_3^{2,2}$ is equal to $T_4^{2,2}$.[3]

Hint: use the tilde form of the twists; and perhaps start by showing that $T_4^{3,3}$ is zero.

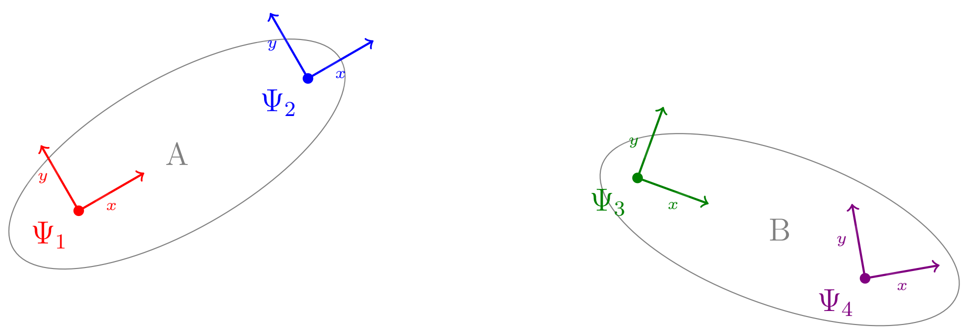

Figure 1: Two bodies with both two frames affixed to it.

[3]Both represent the motion of B w.r.t. A expressed in the same frame, so the result should be the same indeed.

Using the tilde form of the two twists:

$$
\begin{aligned}
\tilde{T}_3^{2,2} &:= \dot{H}_3^2 \cdot H_2^3 \\
&= \dot{H}_3^2 \cdot H_4^3 H_3^4 \cdot H_2^3 \quad \text{(inserting identity)} \\
&= \dot{\overline{H_3^2 H_4^3}} \cdot H_3^4 H_2^3 \quad \text{(taking } H_4^3 \text{ inside time derivative)} \\
&= \dot{H}_4^2 \cdot H_2^4 \\
&= \tilde{T}_4^{2,2}
\end{aligned}
$$

Here we can take H_4^3 inside the time derivative because H_4^3 is constant, so in the product rule for derivation the term $H_3^2 \dot{H}_4^3$ drops out.[4] $\square$

4. Earth and Sun

Let's look at a simple example to apply the theory of twists and H-matrices. Consider a simplified model of the Sun and the Earth, where the Earth is rotating around the Sun in a perfect circle, as if it were fixed to the sun with a rod (see Figure 2).

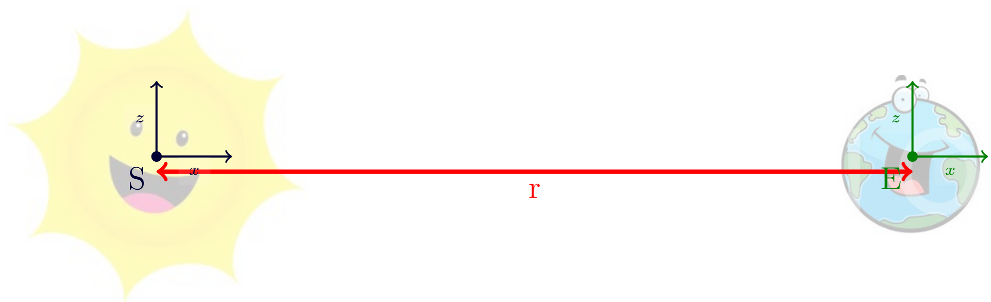

Figure 2: The Sun and the Earth.

Suppose that the motion of the Earth around the Sun is a pure rotation with a velocity of $1\,\mathrm{rad\,s^{-1}}$ around $\hat{z}$, as seen from the Sun. Now say that the Earth is located $2\,\mathrm{m}$ from the Sun[5].

We Earthlings are interested in the speed *we* see, i.e. the same motion but then expressed in *our* frame, so please calculate this velocity.

Hint: Of course you express all velocities as twists, so that you can use H-matrices and Adjoints to express these motions in different frames.

Let's start by writing the H-matrices $H_{\mathrm{e}}^{\mathrm{s}}$ and its inverse, $H_{\mathrm{s}}^{\mathrm{e}}$:

$$
H_{\mathrm{e}}^{\mathrm{s}} = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 2\,\mathrm{m} \\ 0 \\ 0 \end{pmatrix} \\ 0 & 1 \end{pmatrix} \qquad H_{\mathrm{s}}^{\mathrm{e}} = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} -2\,\mathrm{m} \\ 0 \\ 0 \end{pmatrix} \\ 0 & 1 \end{pmatrix}
$$

[4] If you don't see it immediately, you can find this by starting to show that $\tilde{T}_4^{3,3}$ is zero.

[5] It is a shrunken-down, sped-up version of the solar system

The velocity of the Earth in the Sun's frame is a pure rotation around $\hat{z}$, so the Twist is (E w.r.t. S in frame S):

$$T_{\mathrm{e}}^{\mathrm{s,s}} = \begin{pmatrix} 0 \\ 0 \\ 1\,\mathrm{rad\,s^{-1}} \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{2}$$

Twists transform frames using Adjoints, so we have to find the Adjoint of either $H_{\mathrm{s}}^{\mathrm{e}}$ or $H_{\mathrm{e}}^{\mathrm{s}}$. We want to go from frame S to frame E, so we need the Adjoint of the H-matrix that changes coordinates from Ψ_s to Ψ_e—it is $H_{\mathrm{s}}^{\mathrm{e}}$.

$$\begin{aligned} \mathrm{Ad}_{H_{\mathrm{s}}^{\mathrm{e}}} &:= \begin{pmatrix} R_{\mathrm{s}}^{\mathrm{e}} & 0 \\ \tilde{p}_{\mathrm{s}}^{\mathrm{e}} R_{\mathrm{s}}^{\mathrm{e}} & R_{\mathrm{s}}^{\mathrm{e}} \end{pmatrix} \\ &= \begin{pmatrix} I_3 & 0_3 \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & -2 & 0 \end{pmatrix} \cdot I_3 & I_3 \end{pmatrix} \end{aligned}$$

Here you needed the definition of the Adjoint and tilde form; I_3 is the 3×3 identity matrix and 0_3 a 3×3 zero matrix.
We can now easily calculate $T_{\mathrm{e}}^{\mathrm{e,s}}$:

$$\begin{aligned} T_{\mathrm{e}}^{\mathrm{e,s}} &= \mathrm{Ad}_{H_{\mathrm{s}}^{\mathrm{e}}} T_{\mathrm{e}}^{\mathrm{s,s}} \\ &= \begin{pmatrix} I_3 & 0_3 \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & -2 & 0 \end{pmatrix} & I_3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1\,\mathrm{rad\,s^{-1}} \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 1\,\mathrm{rad\,s^{-1}} \\ 0 \\ 2\,\mathrm{m\,s^{-1}} \\ 0 \end{pmatrix} \end{aligned}$$

In other words, Earth travels into the paper ($+\hat{y}$ direction) with a speed of $2\,\mathrm{m\,s^{-1}}$ and also rotates around its $\hat{z}$ axis with a rotational velocity of $1\,\mathrm{rad\,s^{-1}}$.[6]

[6] Note that the latter means that, after a while, the Earth frame has rotated w.r.t. the Sun frame, so $R_{\mathrm{e}}^{\mathrm{s}}$ will not be the identity anymore—we happened to conveniently look at exactly the right time to get an easy calculation.